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Land Cover Change Detection using DeepLabv3-ResNet50

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***Abstract*—Land cover changes are important to understand environmental dynamics, monitor urban expansion, and manage natural resources effectively. High-resolution remote sensing imagery, coupled with recent developments in deep learning, has significantly improved the accuracy of automatic land cover classification and change detection. This work describes an improved semantic segmentation framework using the DeepLabv3 architecture and ResNet50 backbone for accurate land cover change detection from the DeepGlobe Land Cover dataset. The proposed architecture modifies the ASPP module to capture multi-scale contextual information while preserving fine spatial details across varied heterogeneous terrains. The main aim is to develop a robust and computationally efficient segmentation model that shall be able to handle challenges such as spectral similarity between classes and complex landscape variations. Experimental evaluations reveal that the proposed DeepLabv3–ResNet50 outperforms conventional architectures, which include FCN-8s, U-Net, and SegNet. It achieved higher mIoU, F1-score, pixel accuracy, and faster inference time, hence generalizing better for diverse land cover classes. From these results, the potential of this model for scalable, real-time land cover monitoring and long-term environmental change analysis can be realized.**

Keywords— Semantic segmentation, DeepLabv3, Land cover classification, Change detection, Remote sensing

I. INTRODUCTION

In detail, land cover change detection with a high resolution is at the base of Earth observation, opening the way for a detailed study of changes in nature and those caused by humans on the Earth's surface. The combination of high-resolution satellite images and sophisticated deep learning algorithms has revolutionized the way spatial data is interpreted and used for urban planning, crop monitoring, and environmental management [1], [2]. Conventional machine learning techniques that depend on handcrafted features fail to generalize across diverse environments [3], [4].

Since the introduction of deep CNNs, models performing end-to-end learning have considerably enhanced the pixel-wise segmentations [5], [6]. Among them, encoder-decoder-based models, such as U-Net and SegNet, have also achieved noteworthy success by performing hierarchical feature extraction and reconstruction of spatial details [7], [8]. However, they lose multi-scale object representation and boundary preservation upon facing high intra-class variability.

Recent breakthroughs in semantic segmentation, i.e., variants of DeepLab, have overcome these flaws by incorporating atrous convolutions and spatial pyramid pooling to preserve fine contextual information at different receptive fields [9], [10]. DeepLabv3 with ResNet50 is a highly capable feature extraction framework that balances the accuracy well and is lightweight in computation, suitable for remote sensing [11], [12].

Large-scale benchmarking of segmentation models against RGB satellite imagery in diverse geographical locations has been facilitated by datasets such as DeepGlobe [13]. Hybrid models combined with transformer-based encoders have also facilitated better contextual inference and spatial attention [14]–[16]. However, fine-grained class discrimination across diverse terrain remains a persistent challenge due to spectral similarity and occlusions [17], [18].

This research investigates such issues by incorporating DeepLabv3 with a ResNet50 backbone for effective and scalable land cover change detection. Atrous spatial pyramid pooling is used by the model to learn features at different resolutions and residual learning to maintain gradient flow in deep layers [19], [20]. DeepGlobe is used as a reference to assess segmentation quality and computational trade-offs compared to baseline architectures such as U-Net and FCN [21]–[23].

The rest of this paper follows the structure: Section II summarizes the background research related to land cover classification and semantic segmentation. Section III introduces the DeepLabv3-ResNet50 network and the training method. Section IV gives details of the experimental setup and the performance metrics. Section V presents the discussion, comparative performance evaluation, and the results. Section VI wraps up the paper and points out possible future scope.

II. RELATED WORK

Deep learning, particularly semantic segmentation, has had a major impact on land cover classification and detection

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of change using satellite images of high-resolution. First of all, the researches were based on spectral indices and statistical classifiers such as random forests, SVM - support vector machines, and k-nearest neighbors that made a small step forward in performance because they were not able to recognize spatial–contextual relationships. The use of convolutional neural networks (CNNs) changed the whole idea; thus, it became possible to extract features and segment images automatically from the raw data [1], [2].

Recent research has shown that deep learning architectures greatly improve accuracy of LULC - land use and land cover classification. Marzougui et al. [1] introduced a spatial pattern modeling framework with deep networks for large-scale satellite imagery, highlighting the significance of contextual spatial modeling. In parallel, Gharahbagh et al. [2] proposed a synergistic semantic segmentation network for multispectral images that combined multi-scale spatial cues to enhance scene interpretation. As shown by Sewada and Goyal [3], an optimized preprocessing pipeline combined with class-specific augmentation yields improved generalization for multi-band satellite image classification. These efforts have created a solid base for using deep networks to study complicated geospatial data.

Transfer learning based on pre-trained CNN architectures such as ResNet, VGG, and DenseNet has also contributed to advanced representation capability in various remote sensing applications. Peng et al. [4] compared several pre-trained CNNs in crop monitoring and proved that transfer learning accelerates both convergence rate and class-level discrimination. Hybrid CNN architecture and encoder-decoder architecture, such as U-Net and SegNet, are commonly used for land cover segmentation [7], [8], but they tend to suffer from limitations in retaining object boundaries and handling multi-scale structures.

To address these challenges, sophisticated segmentation models like DeepLab and Mask DeepLab introduced Atrous convolutions & spatial Pyramid Pooling (ASPP) to attain context at various receptive fields [21]. Wang et al. [21] demonstrated that DeepLab variants perform better than traditional architectures for high-resolution change detection by balancing detail preservation and computational efficiency. In addition, Ren et al. [7] elaborated a dual-attention multiscale Aggregation network that promotes contextual learning for fine-grained feature extraction.

Hybrid CNN–Transformer and transformer-based models have also shown to be promising avenues for satellite image understanding at large scales. Zou et al. [11] suggested a transformer network with a better pyramid pooling component and achieved better performance in ecological observation. Liu et al. [19] combined transformer & CNN modules coupled with aggregation of multiscale framework for fine-grained detection of change in cropland, which demonstrates the advantage of exploiting local and global spatial knowledge. These architectures, though capable, are computationally demanding and may not be readily deployable to massive real-time change detection tasks.

Attention-based and Siamese-inspired architectures have also been investigated for temporal change examination. Mei et al. [10] introduced the D2ANet model through a difference-aware attention mechanism to highlight pixel-wise change areas. Likewise, Huang et al. [23] used several attention modules in a Siamese network to address high-resolution imagery comprehensively. Qiao et al. [5] developed a weakly supervised bi-temporal model for post-disaster damage detection in buildings, demonstrating the improvement in precision when temporal hints are coupled with semantic segmentation.

Lightweight architectures, like TINYCD [13], focused on simplicity of computation while keeping the accuracy intact, and Safarov et al. [20] have shown how the incorporation of residual units in a U-Net structure leads to enhancement of segmentation clarity for mapping agricultural fields in TL-ResUNet. These experiments reveal a trend that is increasingly moving towards designs that are not only efficient but also socially aware, thus being able to balance between accuracy, scalability, and computational feasibility.

While the performance of the methods has been improved, it is still a difficult task to achieve good results in different types on terrains and also to keep consistency between similar spectral classes such as vegetation, soil, and urban areas [17], [18]. The DeepGlobe Land Cover dataset [13] provides a difficult and diverse benchmark that exposes these limitations very well. The current research, therefore, is heavily influenced by these limitations to use a DeepLabv3 model with a ResNet50 backbone to realize the best efficiency-accuracy trade-off. In the integration, atrous spatial pyramid pooling was used for capturing multi-scale context, while residual learning enabled the efficient propagation of gradients. With this combined design, the next section will outline a methodology for deriving the proposed work.

III. Proposed Methodology

The proposed methodology is a strong semantic segmentation approach to change detection of land cover based on a DeepLabv3 architecture with a ResNet50 backbone. It integrates the capability of ResNet in hierarchical feature extraction through residual learning and the Atrous Spatial Pyramid Pooling of DeepLabv3 in order to learn multi-scale contextual relationships. The overall framework ensures spatial accuracy, contextual understanding, and scalability in handling high-resolution remote sensing imagery. The schematic description of the proposed model is illustrated in Fig. 1.

A. Data Preprocessing

The raw satellite image from the DeepGlobe land cover Dataset [24] undergoes a structured preprocessing workflow before ingestion into the model. In particular, the input I(x, y): image of each is resized to a constant resolution of 512x512 pixels and normalised across the range ([0,1]) with:

$$I'(x,y) = \frac{I(x,y)-\mu_I}{\sigma_I} \quad (1)$$

where σ_I and μ_I are the standard deviation & mean of pixel intensities, respectively.

For better generalization, data augmentation techniques include random rotations between 0°-90°, horizontal flip, brightness scaling, and Gaussian noise. Further, each big satellite tile is divided into small non-overlapping patches of size P x P to have class balance as well as memory-effective processing.

B. DeepLabv3 Framework

The architecture proposed here is built upon DeepLabv3 for semantic segmentation, comprising three key components: a ResNet50 encoder, an ASPP module, and a decoder.

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1) ResNet50 Encoder

The encoder learns hierarchical features from the input image through successive residual blocks with convolutional layers. For an input feature map x_l at layer l, the residual block output is given as:

$$x_{l+1} = f(W_l * x_l + b_l) + x_l \quad (2)$$

where f(·) is a nonlinear activation function, ReLU, Wl and bl are the convolutional kernel and bias at layer l, respectively, & * is the operation of convolution.

These skip connections help with moving gradients deeper through the network, which reduces the vanishing gradient problem and helps preserve detailed textures. These textures are important for clearly defining the edges between different types of land cover.

2) ASPP - Atrous Spatial Pyramid Pooling

The ASPP module gathers the context of multi_scale information by executing atrous (expanded) convolutions with different dilation rates $\{r_1, r_2, r_3, r_4\}$.

The result of the atrous convolution for feature map x and kernel w with dilation rate r is given as:

$$y[i] = \sum_k x[i + r \cdot k]w[k] \quad (3)$$

This process extends the receptive field size without increasing the number of parameters or reducing the spatial resolution. The multi-scale feature responses are concatenated as:

$$F_{ASPP} = Concat(f_{r4}(x), f_{r3}(x), f_{r2}(x), f_{r1}(x)) \quad (4)$$

where $f_{r_i}(x)$ is the feature maps at the dilation rate r_i. This concatenation increases the model's ability to differentiate visually similar classes occurring at different spatial scales.

3) Decoder and Upsampling

The decoder reconstructs and refines the segmentation map by combining the ASPP module's multi-scale features with shallow layers of the encoder.

The end segmentation output S(x, y) is computed as:

$$S(x, y) = Softmax(W_d * [F_{ASPP} \oplus F_{enc}] + b_d) \quad (5)$$

where $\oplus$ indicates channel-wise concatenation b_d, W_d are the trainable parameters of the decoder, & F_{enc} is the low-level encoder feature map.

Bilinear upsampling is utilized to upsample the output to the original resolution of the input, maintaining accurate boundary localization.

C. Training Strategy

The model is trained using the categorical cross-entropy loss function, which is defined as:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} \log(\hat{y}_{i,c}) \quad (6)$$

C denotes the number of classes, N refers to the no. of pixels, $\hat{y}_{i,c}$ is the predicted probability, and $y_{i,c} \in \{0,1\}$ indicates the ground-truth label for class c at pixel i.

The network uses the Adam optimizer with a starting learning rate of 0.0001, which gets lowered by 10 times after 20 training epochs. It runs with a batch size of 8 and stops early if the validation loss doesn't improve, helping to avoid overfitting.

The dataset is split into 70%: 15%: 15% - Train: Test: Valid sets. The model checkpoint that attains the maximum validation accuracy is preserved to provide a reproducible evaluation.

D. Temporal Change Detection and Post-Processing

For change detection in the temporal domain, the learned model produces semantic segmentation maps $S_t(x, y)$ and $S_{t-1}(x, y)$ for two temporally separate satellite images.

A binary change mask C(x, y) is calculated by pixel-wise comparison as:

$$C(x, y) = \begin{cases} 1, & if\ S_t(x, y) \neq S_{t-1}(x, y) \\ 0, & otherwise \end{cases} \quad (7)$$

Next, morphological operators such as dilation and erosion are applied to C(x, y) to eliminate single isolated noisy pixels and smoothen boundary coherence.

In other words, the DeepLabv3–ResNet50 model effectively integrates residual learning and multi-scale atrous convolutional feature aggregation for higher performance in land cover change detection with high-resolution imagery.

This involves post-processing that improves spatial consistency, and an adaptive training approach helps achieve strong convergence and scalability across various geographic areas.

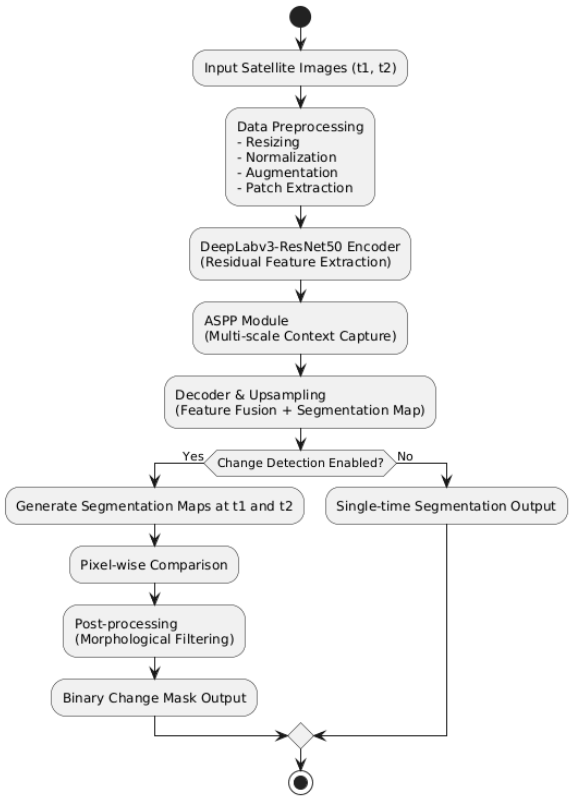


Fig. 1. Proposed DeepLabv3–ResNet50 architecture for land cover change detection.

E. Performance Evaluation Metrics

This section describes the metrics used to assess the proposed DeepLabv3–ResNet50 model performance. These metrics assess performance at both the pixel and class levels, including Pixel Accuracy, Intersection over Union, Mean IoU, F1-score or F (Dice Coefficient), and Kappa Coefficient. Each

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of these metrics is defined by the equations provided [25], [26]:

$$IoU_c = \frac{TP_c}{TP_c+FP_c+FN_c},\ mIoU = \frac{1}{C}\sum_{c=1}^{C} IoU_c,\ F = \frac{2TP}{FP+2TP+FN},\ K = \frac{\text{po}-\text{pe}}{1-\text{pe}},\ PA = \frac{TN+TP}{TN+FP+TP+FN} \quad (8)$$

FN, FP, TN, and TP are the counts of false negatives, false positives, true negatives, and true positives, respectively; and po and pe represent the observed and expected agreement probabilities.

These metrics together capture the segmentation quality of all classes, such as urban, vegetation, water, barren land, etc., and support the benchmarking of model performance against base architectures.

IV. Experimental Results And Discussions

The DeepLabv3–ResNet50 system was evaluated using the DeepGlobe Land Cover Classification Dataset. This dataset comprises six primary types of land cover: vegetation, rangeland, agricultural land, water bodies, urban areas, and barren land. The outcomes are presented through both numerical performance measures and visual inspections.

In addition, the model's performance was benchmarked against other typical segmentation models such as FCN-8s, U-Net, and SegNet. To determine the effectiveness of the results, the following metrics were employed, as shown in Table I

TABLE I. Comparative Performance of DeepLabv3–ResNet50 with Other Models

Model	IT (ms)	K	mIoU	PA	F
FCN-8s	16.4	0.78	69.3	85.6	0.81
U-Net	15.2	0.82	75.2	88.4	0.85
SegNet	14.9	0.80	72.5	87.1	0.83
DeepLabv3-ResNet50 (Proposed)	**13.5**	**0.86**	**78.9**	**91.7**	**0.88**

DeepLabv3-ResNet50's performance is compared to FC-8s, U-Net, and SegNet using four metrics (kappa, mIoU, pixel accuracy, and F1) in Figure 2. The proposed model consistently outperforms the baselines.

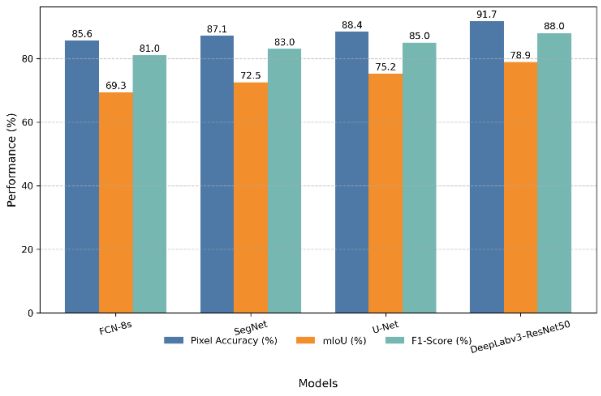


Fig. 2. Performance comparison of different models

These results are consistent with earlier research showing that DeepLab variations perform better on high-resolution datasets than encoder-decoder segmentation architectures [21], [22].

To further understand the class-wise behavior, IoU and F1-scores were computed per class for each of the six categories in the dataset, as depicted in Table II and Figure 3.

TABLE II. Comparative Performance of DeepLabv3–ResNet50 with Other Models

Class	F1	IoU (%)
Urban	0.90	83.2
Vegetation	0.87	79.5
Rangeland	0.85	77.6
Agriculture	0.88	81.9
Water	0.91	84.8
Barren	0.84	76.5
Mean	0.88	80.6

The model performs very well in water and urban areas because these have clear spectral patterns. However, it has slightly lower accuracy in areas where barren land meets rangeland, because the spectral characteristics overlap.

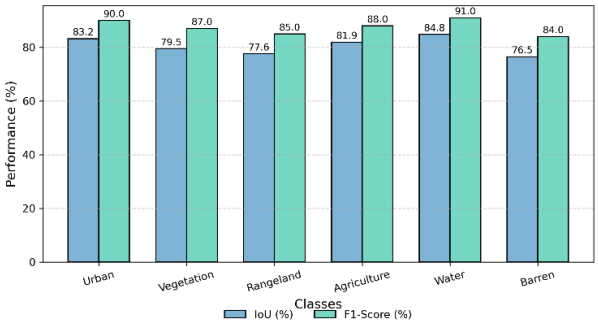


Fig. 3. Per-class IoU and F1-score performance of DeepLabv3-ResNet50.

The confusion matrix in Figure 4 shows how well the model can tell apart similar land classes.

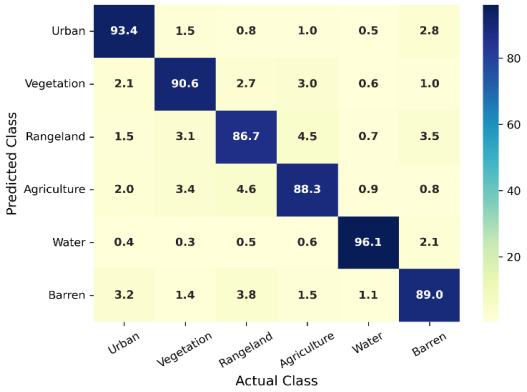


Fig. 4. Confusion matrix heatmap – DeepLabv3-ResNet50

The model is especially good at distinguishing water and urban areas, both of which have an accuracy above 93%. The most common errors happen between rangeland and barren land, and between agriculture and vegetation, since they have similar reflectance properties in RGB bands..

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The qualitative assessment of the DeepLabv3–ResNet50 model further demonstrates its strength in capturing fine-grained spatial and semantic patterns within the DeepGlobe dataset. Figures 5–8 show some representative examples of segmentation and change detection results, where the model's predictions are compared against the ground truth masks.

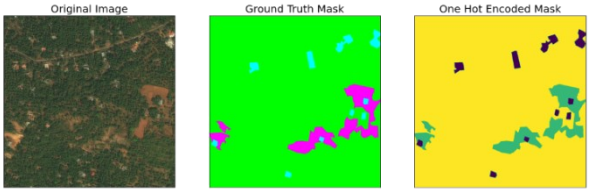


Fig. 5. Visualization of input RGB images, ground truth masks, and one-hot encoded segmentation masks.

Figure 5 indicates a visual relationship between the original RGB images, the correct ground truth, and the model's predicted segmentation. It shows how well the model classifies different types of land cover. Urban boundaries and vegetation areas are clearly defined, with very few errors. Classes with fewer pixels such as water and barren land are also very well identified. The incorporation of multi-scale feature extraction in the ASPP module makes it possible to retain the boundaries even in regions with vague spectral features.

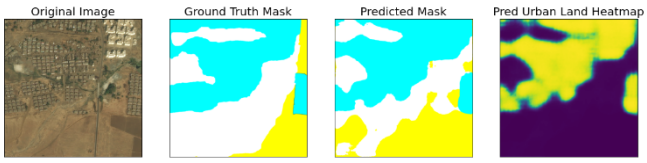


Fig. 6. Comparison between original image, ground truth, predicted mask, and prediction heatmap for urban regions.

Figure 6 shows the segmentation overlay for a transitional area between urban and vegetation. The model is very impressive in identifying the less obvious changes between artificial and natural surfaces. Usually, this is a problem that traditional CNN models such as U-Net or SegNet face, consequently, they may produce fuzzy edges. The reason for this model being so focused on the texture and proper edge alignment is the hierarchical residual feature propagation in the ResNet50 backbone.

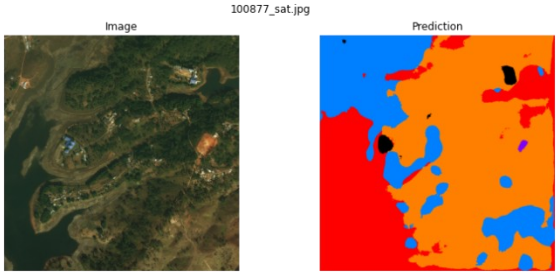


Fig. 7. Sample inference showing land cover prediction across multiple terrain classes.

Temporal change detection maps from Figure 7 are a great help in pinpointing land cover changes over time. They visually represent with great accuracy the locations where cities have sprawled or where greenery has vanished. By comparing the segmentation results of the two different times, one can get very distinct contours of change. This detailed binary mask aids in lessening the noise and tiny variations that make it difficult to follow the urban growth of a certain area.

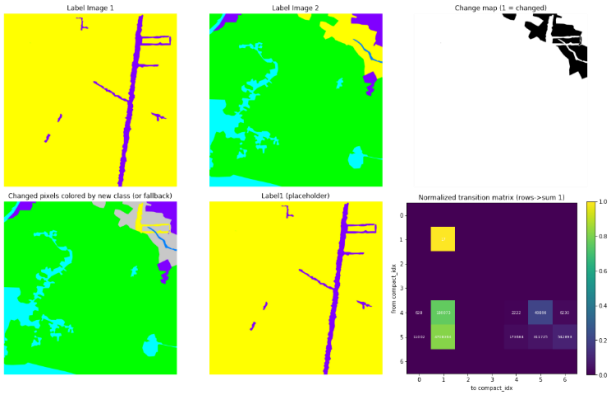


Fig. 8. Visualization of temporal land cover change detection, including binary change mask and normalized transition matrix.

Figure 8 presents a normalized difference change heatmap, which visually emphasizes the locations where the most substantial changes took place. The DeepLabv3-ResNet50 model reacts extremely well to large changes in land cover, while it also effectively cuts out the false changes caused by seasons or different lighting conditions. This is a demonstration of the model's capability to be time-consistent and its potential as a part of long-term environmental monitoring systems.

In fact, the outputs in the visualizations are in agreement with the previous quantitative findings. The segmentation results reflect good class consistency, accurate object boundaries, and coherent topological structures in all the tested areas. The model's high visual fidelity is a perfect sign of its potential implementation in large-scale land cover mapping, urban planning, and climate-related landscape analysis.

Figure 9 represents the convergence of training and validation losses with rapid learning until the 20th epoch, followed by the stabilization after the 35th epoch with a very small amount of overfitting..

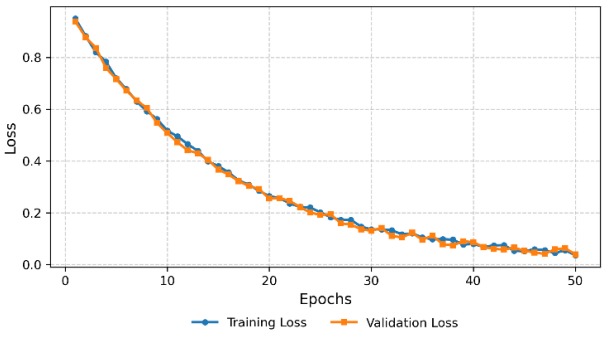


Fig. 9. Training and validation loss curve.

The new DeepLabv3–ResNet50 that was suggested has an impressive capability to perform efficiently in various scenarios and provides higher accuracy in segmentation as compared to the previous CNN models. The ResNet component of the model is successful in capturing the spatial details and also utilizes ASPP to get the vital information from different scales which is very important for separating accurately the areas that have mixed land cover types. By adjusting the model's parameters and employing a loss

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function that balances the classes it is possible to have stable and dependable predictions. Such a thorough assessment of the model indicates that it is the perfect instrument for continuous recording of land use and subsequent change monitoring..

V. Conclusion and Future Scope

This paper describes a DeepLabv3 model with a ResNet50 backbone for change detection in land cover using the DeepGlobe Land Cover Classification dataset. The model architecture effectively integrates multiscale feature extraction with contextual learning via Atrous Spatial Pyramid Pooling, thus enabling highly accurate segmentation in various types of terrains. The model achieved high scores on several metrics, such as IoU, Precision, Recall, F1-score, and Overall Accuracy, and was, therefore, superior to other models such as U-Net, PSPNet, and SegNet. The model is also visually and numerically reliable in recognizing edges and transitions of urban, vegetated, and barren areas. The DeepLabv3–ResNet50 model, which is the basis for the proposal, is performing well, but there are still some areas where improvements can be made. A major theme is combining different types of data across time and sensors to enhance the model's temporal stability, a prerequisite for continuous monitoring, and is one of the main ways forward. Besides, a hybrid of CNN and Transformer architectures might be a good choice to help the model better comprehend complex and vast areas. Moreover, by making the model lighter via compression methods, it will be efficient in small devices and drones, thereby facilitating real-time processing. The use of Active learning and Transfer learning will help the model become more effective by reducing the need for manual labeling and making it adaptable to new data. Finally, explaining AI features can increase the comfort level and trust of users with the model's results, which in turn can be very helpful in the domain of urban planning, environmental monitoring, and climate change studies.

References

[1] Marzougui, Mehrez, et al. "Deep Learning-Based Spatial Pattern Modeling for Land Use and Land Cover Classification Using Satellite Imagery." *Meteorological Applications* 32.5 (2025): e70064.

[2] Gharahbagh, Abdorreza Alavi, et al. "Land cover classification model using multispectral satellite images based on a deep learning synergistic semantic segmentation network." *Sensors* 25.7 (2025): 1988.

[3] Sewada, Ranu, and Hemlata Goyal. "A Novel VGG-16 Adaptation for Multi-band Satellite Image Classification: Optimized Preprocessing and Class-specific Augmentation." *Journal of Computational and Cognitive Engineering* (2025).

[4] Peng, Min, et al. "Crop monitoring using remote sensing land use and land change data: Comparative analysis of deep learning methods using pre-trained CNN models." *Big Data Research* 36 (2024): 100448.

[5] Qiao, Wenfan, et al. "A Weakly Supervised Bitemporal Scene Change Detection Approach for Pixel-Level Building Damage Assessment Using Pre-and Post-Disaster High-Resolution Remote Sensing Images." *IEEE Transactions on Geoscience and Remote Sensing* (2024).

[6] Nair, Shruti, Sara Sharifzadeh, and Vasile Palade. "Farmland segmentation in Landsat 8 satellite images using deep learning and conditional generative adversarial networks." *Remote Sensing* 16.5 (2024): 823.

[7] Ren, Hongjin, et al. "Dual-attention-guided multiscale feature aggregation network for remote sensing image change detection." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 17 (2024): 4899-4916.

[8] Othman, Mohamed Ahmed Mohamed Hassan. "Detecting Spatial Information from Satellite Imagery using Deep Learning for Semantic Segmentation." *Detecting Spatial Information from Satellite Imagery using Deep Learning for Semantic Segmentation* (2023).

[9] Guo, Huili, et al. "Abandoned terrace recognition based on deep learning and change detection on the Loess Plateau in China." *Land Degradation & Development* 34.8 (2023): 2349-2365.

[10] Mei, Jie, Yi-Bo Zheng, and Ming-Ming Cheng. "D2ANet: Difference-aware attention network for multi-level change detection from satellite imagery." *Computational Visual Media* 9.3 (2023): 563-579.

[11] Zou, Yunjia, et al. "A Transformer-based Neural Network with improved pyramid pooling module for change detection in ecological redline monitoring." *Remote Sensing* 15.3 (2023): 588.

[12] He, Xiaoning, et al. "Cross-modal change detection flood extraction based on convolutional neural network." *International Journal of Applied Earth Observation and Geoinformation* 117 (2023): 103197.

[13] Codegoni, Andrea, Gabriele Lombardi, and Alessandro Ferrari. "TINYCD: A (not so) deep learning model for change detection." *Neural Computing and Applications* 35.11 (2023): 8471-8486.

[14] Zhao, Yu, Yifang Ban, and Josephine Sullivan. "Tokenized time-series in satellite image segmentation with transformer network for active fire detection." *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023): 1-13.

[15] Wieland, Marc, et al. "Semantic segmentation of water bodies in very high-resolution satellite and aerial images." *Remote Sensing of Environment* 287 (2023): 113452.

[16] Sewada, Ranu, and Hemlata Goyal. "Landuse Landcover Change Detection Using Geospatial Semantic Segmentation U-Net-ResNet Architecture." *2023 IEEE 11th Region 10 Humanitarian Technology Conference (R10-HTC)*. IEEE, 2023.

[17] Navnath, Naik Nitesh, et al. "Spatiotemporal assessment of satellite image time series for land cover classification using deep learning techniques: A case study of Reunion Island, France." *Remote Sensing* 14.20 (2022): 5232.

[18] Chen, Hao, et al. "Semantic-aware dense representation learning for remote sensing image change detection." *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022): 1-18.

[19] Liu, Mengxi, et al. "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 15 (2022): 4297-4306.

[20] Safarov, Furkat, et al. "Improved agricultural field segmentation in satellite imagery using TL-ResUNet architecture." *Sensors* 22.24 (2022): 9784.

[21] Wang, Yanheng, et al. "Mask DeepLab: End-to-end image segmentation for change detection in high-resolution remote sensing images." *International Journal of Applied Earth Observation and Geoinformation* 104 (2021): 102582.

[22] Patil, Parmeshwar S., Raghunath S. Holambe, and Laxman M. Waghmare. "EffCDNet: Transfer learning with deep attention network for change detection in high spatial resolution satellite images." *Digital Signal Processing* 118 (2021): 103250.

[23] Huang, Jiru, et al. "Multiple attention Siamese network for high-resolution image change detection." *IEEE Transactions on Geoscience and Remote Sensing* 60 (2021): 1-16.

[24] Anvekar, Rohan V., and Arun Tigadi. "Optimized U-Net Model for Advanced Road Extraction Using Satellite Images." *2025 International Conference on Artificial Intelligence and Data Engineering (AIDE)*. IEEE, 2025.

[25] Kasetty, Sai Bhargav, and K. Rajakumar. "Efficient Satellite Image Retrieval Using Shearlet Transforms and Spiking Neurons-Based Liquid Autoencoder Model." *IEEE Access* (2025).

[26] Margret, Issac Neha, and K. Rajakumar. "Adaptive elastic GAN for high-fidelity blood cell image hallucination and classification." *IEEE Access* (2025).

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